

Reza Adousi

Department of Computer Engineering, Khatam University, Tehran, Iran

 (+98) 905-969-0378  rreza.adousi@gmail.com  [RezaAdousi](https://www.linkedin.com/in/RezaAdousi)  [RezaAdousi](https://github.com/RezaAdousi)  rezaadousi.github.io

EDUCATION

Khatam University 2024 – 2028 (Expected)
Bachelor of Science in Computer Engineering — GPA: 18.93/20 Tehran, Iran

- **Relevant Coursework:** Data Structures & Algorithms (19/20), Advanced Programming (20/20), Probability & Statistics (20/20), Artificial Intelligence (20/20), Linear Algebra (20/20), Theory of Languages & Machines (20/20), Differential Equations (19.5/20)

National Organization for Development of Exceptional Talents (NODET) 2021 – 2024
Diploma in Mathematics and Physics — GPA: 18.58/20 Tehran, Iran

RESEARCH INTERESTS

- Multimodal Large Language Models
- Vision-Language Models (VLMs)
- Agentic AI & LLM Agents
- Embodied AI & Interactive Decision Making

PUBLICATIONS

Modeling Player Engagement under Goal Assignment Uncertainty 2026
Under Preparation
R. Adousi, H. Khorram, S. Kamkar

EXPERIENCE

Research Assistant — Human-Computer Interaction & Game AI Research 2026 – Present
Khatam University
Under the supervision of [Dr. Shiva Kamkar](#) Tehran, Iran

- Investigated player behavior and engagement under goal assignment uncertainty by conducting comprehensive literature reviews and synthesizing key methodologies.
- Designed and executed experimental frameworks in interactive game environments to model user responses and validate analytical hypotheses.
- Analyzed experimental telemetry data using statistical approaches to derive quantitative insights for manuscript publication.

Teaching Assistant — Introduction to Computer Science and Programming Spring 2026
Khatam University
Under the supervision of [Prof. Haniyeh Fallah](#) Tehran, Iran

- Led weekly review and Q&A sessions for undergraduate students, reinforcing core Python programming concepts and computer science fundamentals.
- Designed and evaluated coursework assessments, including programming quizzes and debugging assignments, to measure practical problem-solving skills.
- Mentored students in hands-on debugging and algorithm implementation during interactive problem-solving labs.

SELECTED HONORS & AWARDS

2024	FANAP Distinguished Scholarship for Top-Ranked Students	\$1,200/year (for 4 years)
2026	Top 3 Computer Engineering Student Each Semester	Among 70 students
2025	9th Place, “Algorithm LevelUp” — Quera, Algorithmic Contest (C++)	135 participants
2023	Top 2%, University Entrance Exam	160,000+ participants
2022	Iranian National Computer Olympiad, Second-Round Qualifier	Top 0.2% of 10,000

SELECTED PROJECTS

Neural Networks from Scratch <i>Python, PyTorch, NumPy, Git</i>	Summer 2026
- Implemented foundational deep learning architectures and components (Autograd engine, MLP, CNN, ResNet, Transformer) line-by-line from scratch to solidify core mathematical and operational principles of deep learning. - Translated core concepts from key research papers and Andrej Karpathy's lectures into clean, production-ready code, covering backpropagation, attention mechanisms, vector embeddings, and manual gradient tracking. - Benchmarked custom implementations against PyTorch primitives to evaluate numerical accuracy, memory consumption, training stability, and performance trade-offs.	
Scientific Paper Analyzer <i>Google ADK, Gemini API, MCP, Python, Streamlit</i>	Summer 2026
- Built an end-to-end multi-agent workflow using Google ADK and Gemini API to process scientific PDFs, extract structured insights, and generate Markdown research reports. - Implemented a human-in-the-loop (HITL) approval mechanism enabling iterative agent refinement based on user feedback. - Integrated a custom Model Context Protocol (MCP) server for local file I/O, including PDF parsing with PyMuPDF and local report generation. - Designed a multi-layer security checkpoint to detect prompt injection attempts, redact PII, and block potentially malicious inputs before model execution.	
RateMyProf <i>Python, Flask, SQLAlchemy, Hugging Face, Jinja2</i>	Winter 2025
- Architected and developed a full-stack web application enabling university students to evaluate faculty members, with secure authentication and relational database modeling. - Integrated Hugging Face NLP models to perform sentiment analysis and automatically synthesize student feedback into concise summaries. - Implemented business logic and database constraints to enforce single-review limits per user/professor and dynamically update summary insights upon review submission or deletion.	
Sentiment Analysis & Hybrid Stacking Ensemble <i>Python, Scikit-learn, PyTorch, Hugging Face</i>	Fall 2025
- Engineered a hybrid stacking ensemble combining Logistic Regression, Linear SVM, and DistilBERT, achieving 91.6% accuracy on 525K+ Amazon Fine Food Reviews. - Preprocessed textual data with TF-IDF and trained classical ML baselines, with Logistic Regression and Linear SVM achieving approximately 89.8% accuracy. - Addressed Transformer sequence-length constraints by filtering inputs to ≤ 512 tokens and combining model predictions through a Logistic Regression meta-model.	

SELECTED CERTIFICATES & COURSES

Machine Learning Specialization — DeepLearning.AI / Stanford Online	2026
Neural Networks: Zero to Hero — Andrej Karpathy (Self-Study)	2026
CS231n: Deep Learning for Computer Vision — Stanford University (Self-Study)	2026
CS50x: Introduction to Computer Science — Harvard University	2025
IAAA: Iran's Annual Artificial Intelligence Award — AI Program	2025
AI & Data Engineering — FANAP Campus	2024

TECHNICAL SKILLS

Programming & ML: Python, C++, PyTorch, TensorFlow, Scikit-learn, OpenCV, Pandas, NumPy
AI & LLMs: Hugging Face Transformers, LLMs, Agentic AI, Google ADK, MCP
Web Development: HTML/CSS, JavaScript, Flask, SQLAlchemy, Jinja2
Research: Study Design, Scientific Writing, Statistical Analysis, Data Preprocessing
Tools: Git, GitHub, Linux, LaTeX
Languages: Persian (Native), English (Advanced), German (Intermediate)